Nathan Soldati

Zurich, Switzerland — nathan.soldati01@bluewin.ch — LinkedIn — Portfolio & Projects

EDUCATION

ETH Zurich – MSc Information Technology and Electrical Engineering *Sep 2024 – Feb 2027*

- Focus: Machine Learning, AI, Computer Vision, Automation, and Optimization.
- Master thesis (ongoing): 4D Computer Vision, collaboration between ETH CVG (Pollefeys) and Google Zürich (Tombari).

University of Melbourne – Exchange Semester *Feb 2025 – Jun 2025*

ETH Zurich – BSc Information Technology and Electrical Engineering *Sep 2021 – Aug 2024*

- GPA: 5.88/6.00, top 2%.
- Bachelor thesis: Multi-Layer Feedback Control for Selective Laser Melting Using a Backpropagation Strategy.

PROFESSIONAL EXPERIENCE

AI & Computer Vision Engineer – Swiss Engineering Partners AG *Feb 2026 – Aug 2026*

- Developed a multi-stage object detection system using RT-DETR and DINOv3 backbones for detecting safety-relevant objects in industrial video.
- Built the end-to-end data pipeline, including frame selection, VLM-assisted pseudo-labeling, bounding-box annotation, dataset curation, training, and evaluation.
- Evaluated both cloud and NVIDIA Jetson edge deployment.

Machine Learning Engineer – Amstein+Walthert *Sep 2025 – Dec 2025*

- Fine-tuned ViTs on 1,000+ architectural floor plans for semantic segmentation.
- Built ML workflows for extracting structured information from complex visual design documents.

RESEARCH & TEACHING EXPERIENCE

Teaching Assistant – ETH Zurich *Sep 2022 – Sep 2024*

- Taught weekly tutorials and graded coursework for 120+ undergraduate students across 3 core subjects.

Research Assistant – Automatic Control Laboratory, ETH Zurich *Feb 2024 – Aug 2024*

- Developed a control framework for stabilizing a cart-pole system for a new course.

TECHNICAL SKILLS

Machine Learning: PyTorch, Hugging Face, OpenCV, VLMs, Diffusion Models

Systems / Deployment: CUDA, ONNX, TensorRT, Distributed Training, Docker, Linux, Git

Programming: Python, C/C++, MATLAB/Simulink

Languages: Italian (Native), English and German (Fluent), French (Conversational)

AWARDS

- **Nizzola Excellence Scholarship:** Awarded for exceptional academic performance during bachelor's studies.
- **ITET Best BSc Thesis Award:** Selected among the department's top three BSc theses.